

Newton's inscribed polygon differs from the parabola by the phase of its segments

For a body of mass m under a constant transverse force F on $[0, \tau]$, take any partition into steps $\tau_1, \dots, \tau_N$ and Newton's polygon of Proposition I inscribed in the Galileo parabola: uniform motion along each chord, impulses at the vertices. Classically the polygon and the curve are in the same state at every vertex. Quantum mechanically their evolutions differ by a pure phase, the same for every state of the body:

$$W_{\text{poly}} = e^{i\theta_N} W_F, \quad \theta_N = \frac{F^2}{24m\hbar} \sum_{j=1}^N \tau_j^3 = \frac{F}{2v\hbar} \sum_{j=1}^N S_j,$$

where $S_j = vF\tau_j^3/(12m)$ is the parabolic segment cut off by the j -th chord at horizontal speed v , half of Lemma XI's tangent area for that step. Three consequences follow.

- **Insertion.** Inserting a vertex at fraction λ of a step of duration τ_j lowers the phase by $F^2\tau_j^3\lambda(1-\lambda)/(8m\hbar)$, which is $F/(2v\hbar)$ times the triangle the new vertex inscribes. At $\lambda = \frac{1}{2}$ this is Archimedes' triangle, three quarters of the segment, so repeated halving removes the phase in the proportions of his exhaustion, $1 : \frac{1}{4} : \frac{1}{16} : \dots$.
- **Ordering.** The same Weyl increments composed in reverse order give the phase $\Phi_N = F^2(\tau^3 - \sum_j \tau_j^3)/(6m\hbar)$ relative to the chronological order, the largest over all orderings.
- **Conservation.** For every partition,

$$\Phi_N + 4\theta_N = \frac{F^2\tau^3}{6m\hbar} = \frac{\tau\Delta E}{3\hbar} = \frac{F}{v\hbar} A_{\text{inertial,fall}},$$

so refinement moves phase from the polygon's segments into the ordering holonomy and keeps the total fixed at the Galileo area over $\hbar$.

Both phases are c-numbers, so squeezing, mixing or any other choice of body state leaves them unchanged. That is the property the two quantum floors lacked, as corrected on 2026-09-22 in the mark-cost and probabilistic notes. The single-step case is the chord lens of the constant-force note, and the closed loop of N01 is another instance of Lemma 2 below. New here are the identity for every partition, the insertion law, the ordering holonomy and the conservation law. Exploratory; no ledger promotion.

1. Setting and the exact lift

Transverse coordinate $\hat{y}$, momentum $\hat{p}$, $[\hat{y}, \hat{p}] = i\hbar$, mass $m > 0$; the horizontal motion at speed v is common to every history considered and cancels. A force history is an impulse density or measure f on $[0, \tau]$ with Hamiltonian $H_f(t) = \hat{p}^2/2m - f(t)\hat{y}$. In the interaction picture of the free evolution U_0 , write $W_f(t) = U_0(t)^\dagger U_f(t)$ and use the Weyl operators of the probabilistic note,

$$D(a, b) = \exp\left[\frac{i}{\hbar}(b\hat{y} - a\hat{p})\right], \quad D(z_1)D(z_2) = e^{-i\omega(z_1, z_2)/2\hbar} D(z_1 + z_2),$$

with $\omega(z_1, z_2) = a_1b_2 - a_2b_1$. An impulse J at time t is $U_0(t)^\dagger e^{iJ\hat{y}/\hbar} U_0(t) = D(-Jt/m, J)$, since $U_0(t)^\dagger \hat{y} U_0(t) = \hat{y} + \hat{p}t/m$. The history's **phase-plane path** is

$$z_f(t) = (\alpha_f(t), \beta_f(t)) = \left(-\frac{1}{m} \int_0^t s f(s) ds, \int_0^t f(s) ds\right),$$

the accumulated transverse displacement relative to free motion and the accumulated impulse. For the constant force, $z_F(t) = (-Ft^2/2m, Ft)$, a parabola in the phase plane.

Lemma 1. $W_f(\tau) = \exp[\frac{i}{2\hbar} \int_0^\tau \omega(z_f, dz_f)] D(z_f(\tau))$.

Proof. For finitely many impulses with increments $\delta z_1, \dots, \delta z_N$ in time order, induction on the Weyl relation gives $D(\delta z_N) \cdots D(\delta z_1) = \exp[\frac{i}{2\hbar} \sum_{i < j} \omega(\delta z_i, \delta z_j)] D(\sum_j \delta z_j)$, and $\sum_{i < j} \omega(\delta z_i, \delta z_j) = \sum_j \omega(z_{j-1}, \delta z_j)$ with $z_{j-1} = \sum_{i < j} \delta z_i$. For a density f the generator $\frac{i}{\hbar} f(t)(\hat{y} + \hat{p}t/m)$ has c-number commutators at different times, so the Magnus series stops at second order and the Riemann sums converge to $\int \omega(z_f, dz_f)$, as in N01 §4. $\square$

Lemma 2. If $z_f(\tau) = z_g(\tau)$, then $W_f W_g^\dagger = e^{i\mathcal{A}(f,g)/\hbar} \mathbf{1}$, where $\mathcal{A}(f,g) = \frac{1}{2} \oint \omega(z, dz)$ is the signed area enclosed by z_f followed by z_g reversed.

Proof. Lemma 1 for both, and Green's theorem for the closed loop. $\square$

Two histories with equal endpoints leave the body in the same classical state after τ , whatever happened in between, and Newton's mechanics has no later observation that separates them. Lemma 2 says the quantum evolutions still differ, by the enclosed phase-plane area over $\hbar$, and that the difference is a multiple of the identity. This is the standard geometric phase of displaced oscillators, used in trapped-ion phase gates (Sørensen and Mølmer, PRA **62**, 022311, 2000; Leibfried et al., Nature **422**, 412, 2003, both metadata). The t^3 phase of a freely falling wave packet is the single-history form of it (Greenberger and Overhauser, RMP **51**, 43, 1979, metadata).

2. The inscribed polygon

Partition $0 = t_0 < \cdots < t_N = \tau$ with $\tau_j = t_j - t_{j-1}$ and midpoints $c_j = (t_{j-1} + t_j)/2$. **Newton's inscribed polygon** gives the impulse $F\tau_1/2$ at t_0 , $F(\tau_j + \tau_{j+1})/2$ at each interior t_j , and $F\tau_N/2$ at t_N : each step's impulse $F\tau_j$ is split equally between its two ends.

Theorem 1. (i) On (t_{j-1}, t_j) the polygon moves uniformly with momentum Fc_j along the chord of the parabola, and its phase-plane path passes through $z_F(t_j)$ at every vertex. (ii) $W_{\text{poly}}(\tau) = e^{i\theta_N} W_F(\tau)$ with $\theta_N = F^2 \sum_j \tau_j^3 / (24m\hbar)$. (iii) With $S_j = vF\tau_j^3 / (12m)$ the area between the j -th arc and its chord, $\theta_N = (F/2v\hbar) \sum_j S_j$.

Proof. (i) After the second half of the impulse at t_{j-1} the momentum is $Ft_{j-1} + F\tau_j/2 = Fc_j$, and in time τ_j it carries the body through $Fc_j\tau_j/m = F(t_j^2 - t_{j-1}^2)/2m$, the parabola's transverse increment. In the phase plane the two half-impulses of step j are $k_1 = (-F\tau_j t_{j-1}/2m, F\tau_j/2)$ and $k_2 = (-F\tau_j t_j/2m, F\tau_j/2)$, whose sum is the parabola's increment $\delta z_j = (-F\tau_j c_j/m, F\tau_j)$.

(ii) Both evolutions are products over steps, in the same order, of operators with the same displacements δz_j , so their ratio is the product of the per-step phase ratios. By Lemma 1 the polygon's step is $D(k_2)D(k_1) = e^{i\omega(k_1, k_2)/2\hbar} D(\delta z_j)$ with $\omega(k_1, k_2) = \frac{F\tau_j}{2} \cdot \frac{F\tau_j}{2m} (t_j - t_{j-1}) = \frac{F^2\tau_j^3}{4m}$. The curve's step, with $s = t - t_{j-1}$ and path $(-F(s^2 + 2st_{j-1})/2m, Fs)$ from $z_F(t_{j-1})$, has $\omega(z, dz) = \frac{F^2 s^2}{2m} ds$, hence phase $\frac{1}{2\hbar} \int_0^{\tau_j} \frac{F^2 s^2}{2m} ds = \frac{F^2\tau_j^3}{12m\hbar}$. Neither depends on t_{j-1} . The ratio is $\exp[\frac{i}{\hbar}(\frac{1}{8} - \frac{1}{12})\frac{F^2\tau_j^3}{m}] = \exp[\frac{iF^2\tau_j^3}{24m\hbar}]$.

(iii) Over the step, the chord exceeds the parabola by $\frac{F}{2m}(t - t_{j-1})(t_j - t)$, and integrating against $dx = v dt$ gives $S_j = vF\tau_j^3/12m$. Then $\frac{F}{2v} S_j = \frac{F^2\tau_j^3}{24m\hbar}$. $\square$

Lemma XI's tangent area for the step is $A_j = vF\tau_j^3/6m = 2S_j$, so $\theta_N = (F/4v\hbar) \sum_j A_j$, and Proposition 1 of the paper is the statement that this sum vanishes with the mesh.

Corollary 2 (insertion). Replacing step j by steps $\lambda\tau_j$ and $(1 - \lambda)\tau_j$ lowers θ_N by

$$\frac{F^2\tau_j^3}{24m\hbar} [1 - \lambda^3 - (1 - \lambda)^3] = \frac{F^2\tau_j^3 \lambda(1 - \lambda)}{8m\hbar} = \frac{F}{2v\hbar} T_j(\lambda),$$

with $T_j(\lambda) = S_j - S'_j - S''_j$ the triangle inscribed between the old chord and the new vertex. At $\lambda = \frac{1}{2}$, $T_j = \frac{3}{4}S_j$, which is Archimedes' *Quadrature of the Parabola*, Propositions 17 and 24 (the segment is four thirds of its inscribed triangle, by balancing and by exhaustion; companion). After k rounds of halving every step, θ is 4^{-k} of its value. $\square$

Each inserted point therefore changes the quantum evolution by a computable phase, fixed by the area the insertion removes, and the change is the same for every state of the body. A step's own phase reaches one radian at

$$\tau_h = \left(\frac{24m\hbar}{F^2} \right)^{1/3},$$

the scale that the constant-force note found for the single chord lens. The insertion mesh of the mark theorem, $\tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$, has the same form, obtained there from the cost of a mark and here from the phase of a segment with no apparatus at all.

3. The ordering holonomy and the conservation law

Keep the curve's step operators $V_j = e^{i\varphi_j} D(\delta z_j)$ and compose them in another order. For a permutation σ , write U_σ for the product with the steps applied in the order σ .

Theorem 3. $U_{\text{chron}} U_\sigma^\dagger = \exp[\frac{i}{\hbar} \sum \omega(\delta z_i, \delta z_j)] \mathbf{1}$, the sum over the pairs $i < j$ that σ inverts. Every term is positive,

$$\omega(\delta z_i, \delta z_j) = \frac{F^2}{m} \tau_i \tau_j (c_j - c_i) > 0 \quad (i < j),$$

so the reversed order is extremal, with

$$\Phi_N = \frac{1}{\hbar} \sum_{i < j} \omega(\delta z_i, \delta z_j) = \frac{F^2}{6m\hbar} \left(\tau^3 - \sum_j \tau_j^3 \right).$$

Proof. The per-step phases φ_j are common to every ordering. By the Weyl relation the phase of a product is $\frac{1}{2\hbar}$ times the sum of $\omega(\delta z_a, \delta z_b)$ over pairs with a applied before b ; a pair kept in order cancels between the two products, and an inverted pair contributes $\omega(\delta z_i, \delta z_j) - \omega(\delta z_j, \delta z_i) = 2\omega(\delta z_i, \delta z_j)$. With $\delta z_i = (-F\tau_i c_i/m, F\tau_i)$, $\omega(\delta z_i, \delta z_j) = \frac{F^2}{m} \tau_i \tau_j (c_j - c_i)$. For the sum, $c_j - c_i = \frac{\tau_i}{2} + \sum_{i < k < j} \tau_k + \frac{\tau_j}{2}$, so $\sum_{i < j} \tau_i \tau_j (c_j - c_i) = \frac{1}{2} \sum_{i \neq j} \tau_i^2 \tau_j + \sum_{i < k < j} \tau_i \tau_k \tau_j$, and $(\sum_j \tau_j)^3 = \sum_j \tau_j^3 + 3 \sum_{i \neq j} \tau_i^2 \tau_j + 6 \sum_{i < k < j} \tau_i \tau_k \tau_j$ gives $\frac{1}{6}(\tau^3 - \sum_j \tau_j^3)$. $\square$

Corollary 4 (conservation). $\Phi_N + 4\theta_N = F^2 \tau^3 / (6m\hbar)$ for every partition, and $F^2 \tau^3 / 6m = \tau \Delta E / 3 = (F/v) A_{\text{inertial,fall}}$ with $A_{\text{inertial,fall}} = v F \tau^3 / 6m$ the paper's area (1). $\square$

In areas, $\hbar \Phi_N = (F/v)(A - \sum_j A_j)$ and $4\hbar \theta_N = (F/v) \sum_j A_j$: the ordering holonomy carries the part of the Galileo area that refinement keeps, the polygon defect carries the part that Proposition 1 sends to zero, and the weighted total is fixed. For the uniform partition, $\Phi_N = \frac{\tau \Delta E}{3\hbar} (1 - N^{-2})$.

The chronological and reversed products realize the same displacement, so the holonomy is a property of the composition law. Realizing a reversed order in the laboratory needs displacements as well as impulses, since an impulse applied at the wrong time must be translated back; trapped-ion practice supplies such operations. Theorem 1 needs only two force histories that act on a real body.

4. What the phases give as a record

Put the two histories of Theorem 1 on the arms of a control qubit, as in N01 §4. The body ends in the same state on both arms, so the qubit acquires the relative phase θ_N and nothing else, and the optimal single-shot error in deciding polygon against curve is $\epsilon = \frac{1}{2}(1 - |\sin(\theta_N/2)|)$, with N01's formula for n copies. A single step is resolved from its chord at error ϵ only if

$$\frac{F^2 \tau_j^3}{24m\hbar} \geq 2 \arcsin(1 - 2\epsilon), \quad \tau_j \geq \left(\frac{48 m \hbar \arcsin(1 - 2\epsilon)}{F^2} \right)^{1/3},$$

for every state of the body, pure or mixed, squeezed or not. The bound concerns this readout: the body's own displacement is a separate record, priced by the mark and aperture theorems, and squeezing acts on that record. What the phase supplies is the part of the signal that no preparation can move.

5. The premise, restated

The identities above use the Weyl relations and nothing else: no state, no variance, no apparatus. Of the two quantum premises used so far in this programme, Robertson's inequality is the variance consequence of the commutator, and the 2026-09-22 corrections show that squeezing moves it. The commutator's phase content is what Theorems 1 and 3 use, and squeezing leaves it fixed. So the premise that carries $\hbar > 0$ into the Galileo comparison without a resource bound is the **central extension**: an impulse J and a displacement d compose, in Newton's Corollary I, only up to the phase $Jd/\hbar$ of their parallelogram. Classically Corollary I is exact and commutative, the polygon and the curve are the same state at every vertex, and every θ_N vanishes.

Newton's fits supply a Newton-age form of that premise. Two passages of the *Opticks*, now held verbatim in the companion, fix how the interval of the fits varies with the corpuscle's speed. Book II Part III Prop. XVII, "manifest by the 10th Observation", makes the intervals in two mediums "as the Sine of Incidence to the Sine of Refraction", so the interval is shorter in the denser medium by the refractive ratio. Prop. X takes light to be "swifter in Bodies than in Vacuo, in the proportion of the Sines", the hypothesis of Newton's emission theory. Together they give

$$\Lambda v = \text{const across media}, \quad \text{hence} \quad \Lambda p = \text{const across media}$$

for each colour, the corpuscle's mass being unchanged by refraction.

Proposition 5. Suppose, extending Props. X and XVII from a step in the medium to a continuously varying speed, that a corpuscle's fits advance by one interval $\Lambda = \Lambda_0 p_0/|p|$ per length of path. Then the number of fits along a path is its Maupertuis action $W = \int p \cdot dq$ in units of $\Lambda_0 p_0$. For the inscribed polygon and the curve of Theorem 1 the counts differ by

$$\frac{\Delta W}{\Lambda_0 p_0} = \frac{F^2}{12m\Lambda_0 p_0} \sum_j \tau_j^3 = \frac{1}{2\Lambda_0 p_0} \frac{F}{v} \sum_j A_j$$

fits, with A_j Lemma XI's tangent areas.

Proof. Along the actual motion $p \parallel dq$, so $|p| ds = p \cdot dq$ and the count is $\int p \cdot dq / (\Lambda_0 p_0)$. The horizontal parts are common. On step j the curve has $\int p_y dy = \frac{F^2}{m} \int_{t_{j-1}}^{t_j} t^2 dt$ and the chord has $\frac{F^2}{m} c_j^2 \tau_j$; their difference is $\frac{F^2}{m} \tau_j [\frac{1}{3}(t_{j-1}^2 + t_{j-1}t_j + t_j^2) - \frac{1}{4}(t_{j-1} + t_j)^2] = \frac{F^2 \tau_j^3}{12m}$, and $F^2 \tau_j^3 / 6m = (F/v) A_j$. $\square$

In Newton's optics the fit at arrival decides whether the ray is reflected or transmitted at the next surface (Prop. XII), so on his own terms the polygon and the curve are optically different corpuscles once their fits counts differ by half an interval, although they agree in position and velocity at every vertex. Every ingredient is his: Lemma XI's areas, the measured $\Lambda = 1/89000$ inch, the refraction invariance of Λp from Props. X and XVII, and the fits as a determinate periodic disposition. The unknown is p , which he could not measure. Against the quantum statement, with $\Lambda = \lambda/2$ and $\lambda p = 2\pi\hbar$, the fits phase in radians is $4\theta_N$: one factor 2 because Newton's interval is half a wavelength, and one because $\Delta W = 2\Delta S$ for this pair of histories.

The determinism of Prop. XII is untouched by this; what the fits supply is a phase whose rate is proportional to momentum, and that is the content of the central extension. So the premise that carries $h > 0$ in the squeeze-immune form is present in Newton's optics, with Λp a positive refraction invariant. Two limits remain. Prop. XV's rule for oblique emergence, a product of two secants, is more complicated than a count of intervals along the path, so Proposition 5's extension is ours. And Λp is a per-colour constant in Newton's system: universality across colours is what Planck's constant adds.

6. Every force law, and one functional for both legs

Let f be any integrable force history on an interval of duration τ , and compare the motion with its **chord**, the uniform motion between the same two positions. The deviation $\delta y = y_{\text{chord}} - y_f$ obeys $m \delta y'' = -f$ with $\delta y = 0$ at both ends, so $\delta y = G_\tau f/m$ with the Dirichlet Green's function $G_\tau(s, u) = \min(s, u)(\tau - \max(s, u))/\tau$ in time measured from the start of the interval. Define the **kinetic action of the deviation**

$$\mathcal{K}_\tau[f] = \int_0^\tau \frac{1}{2} m \delta \dot{y}^2 dt = \frac{1}{2m} \int_0^\tau \int_0^\tau G_\tau(s, u) f(s) f(u) ds du \geq 0,$$

zero only for $f = 0$. For the constant force it is $F^2 \tau^3 / 24m = \tau \Delta E / 12 = (F/2v)S$, with S the segment of the chord.

Theorem 7. (a) On each step of a partition, the two impulses $J_j^- = \int (1 - s/\tau_j) f$ and $J_j^+ = \int (s/\tau_j) f$ at its ends reproduce the step's impulse and displacement, so Newton's polygon built from them agrees with the motion at every vertex, and

$$W_{\text{poly}} = \exp \left[\frac{i}{\hbar} \sum_j \mathcal{K}_{\tau_j}[f] \right] W_f.$$

Inserting vertices never raises the phase: $\sum_j \mathcal{K}_{\tau_j}[f] \leq \mathcal{K}_\tau[f]$. (b) For every protocol of marks with uncorrelated error and recoil and $\delta_j \Delta_j \geq \kappa$, with the initial position and velocity unknown, the deflection between the motion under f and free motion obeys

$$d^2 \leq \frac{\mathcal{K}_\tau[f]}{\kappa},$$

with equality approached by dense protocols when f has one sign. (c) Hence at $\kappa = \hbar/2$ the best deflection any such protocol can reach is **twice the phase between the motion and its chord**, $d_{\text{max}}^2 = 2\mathcal{K}_\tau/\hbar = 2\theta_1$.

Proof. (a) Measure time from the start of the step. The impulses at 0 and τ_j are $k^- = (0, J^-)$ and $k^+ = (-\tau_j J^+/m, J^+)$ in the phase plane, and $J^- + J^+ = \int f$, $\tau_j J^+ = \int s f$ match the step's increment. Lemma 1 gives the polygon's step phase $\frac{1}{2\hbar} \omega(k^-, k^+) = \tau_j J^+ J^- / 2m\hbar$ and the motion's step phase $\frac{1}{2\hbar} \int \omega(z, dz) = \frac{1}{2m\hbar} \iint_{u < s} (s - u) f(s) f(u)$, because along the path $\omega(z, dz) = f(s) \frac{1}{m} \int_0^s (s - u) f(u) du ds$. Now $\tau_j J^+ J^- = \iint s(\tau_j - u) f(s) f(u) / \tau_j$; symmetrizing and subtracting $\frac{1}{2} \iint |s - u| f f$ leaves the kernel $\frac{1}{2\tau_j} [s(\tau_j - u) + u(\tau_j - s) - \tau_j |s - u|] = G_{\tau_j}(s, u)$. Shifting the time origin is the shear $(a, b) \mapsto (a - t_{j-1} b/m, b)$, which preserves ω , so the per-step phases are $\mathcal{K}_{\tau_j}/\hbar$, and they multiply as in Theorem 1. The Dirichlet principle $2m\mathcal{K}_\tau = \sup_T (2 \int f T - \int T'^2)$ over T vanishing at the ends, restricted to T vanishing also at the interior vertices, gives the monotonicity.

(b) As in Theorem 2 of the paper, $u^\top \Sigma u \geq 2\kappa E/m$ and $u^\top P = \frac{1}{m} \int_0^\tau f T$ with T vanishing at both ends. Since $-(G_\tau f)'' = f$, $\int f T = \int (G_\tau f)' T' \leq \sqrt{\langle f, G_\tau f \rangle} \sqrt{E}$, so $d^2 \leq \langle f, G_\tau f \rangle / (2m\kappa) = \mathcal{K}_\tau / \kappa$. For f of one sign take $T \propto G_\tau f$, which has one sign while T'' has the other, so the interior weights and the S_j have opposite signs throughout and the dense balanced construction of Theorem 2 applies.

(c) Theorem 1 with a single step is the chord, whose phase is $\theta_1 = \mathcal{K}_\tau/\hbar$. $\square$

The two legs of the programme are governed by one functional of the force history. $\mathcal{K}_\tau$ is what Newton's refinement removes: it is the kinetic action of the motion relative to its chord, and for a constant force it is the chord's segment times $F/2v$. Planck's constant converts it into the phase by which the motion differs from its inscribed polygon, a c-number that no preparation moves; the mark cost converts it into the largest statistical distance any record of the motion can reach. At $\kappa = \hbar/2$ the second is exactly twice the first. The 24 in Theorem 1's $F^2\tau^3/24m$ and in the sharp bound $d^2 \leq F^2\tau^3/24m\kappa$ of the paper's Theorem 2 is one integral, $\iint G_\tau = \tau^3/12$, multiplied by $F^2/2m$.

7. Consequence for STATE

STATE's next item 1 is discharged in a stronger form than it was stated. The state-independent quantity is identified for every refinement: Newton's inscribed polygon differs from the parabola by $\theta_N = (F/2v\hbar)\sum_j S_j$, each inserted vertex changes it by the inscribed triangle over $\hbar$ in Archimedes' proportions, and the ordering holonomy carries the complement, with $\Phi_N + 4\theta_N = \tau\Delta E/3\hbar$. The Newton-age form of the central extension is in §5: Props. X and XVII make Λp a refraction invariant, and the fits count separates the polygon from the curve by $(F/v)\sum_j A_j/(2\Lambda p)$. Section 6 settles the general force law and joins the legs: for any force history the polygon phase is $\sum_j \mathcal{K}_{\tau_j}/\hbar$, the sharp mark bound is $d^2 \leq \mathcal{K}_\tau/\kappa$, and at $\kappa = \hbar/2$ the best recorded deflection is twice the chord phase.